

Natural Resources Price Modeling

using Time Series Methodology

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1. Motivation

- ▶ Henry Hub Natural Gas Price prediction (weekly price, delivery in Erath, Louisiana)

2. Data

- ▶ Endogenous (Crude Oil weekly price– WTI – Cushing, Oklahoma)
- ▶ Exogenous (weather and storage – HDD, HDDdev, CDD, CDDdev, STOR_ch, HH_shutin)

3. Methodology

- ▶ ARIMA, ARIMAX, Transfer Function Model, ECM

4. Results

5. Conclusions

- ▶ Short time horizon prediction – ARIMA, ARIMAX, Transfer Function Model
- ▶ Long time horizon prediction – ECM

Motivation

- ▶ Natural resources – mostly are limited and it is important how we are going to use them
- ▶ Prime usage of natural resources - ENERGY
- ▶ The biggest energy consumer in the world is North America (consumes one-quarter of the world's primary energy production)

Motivation

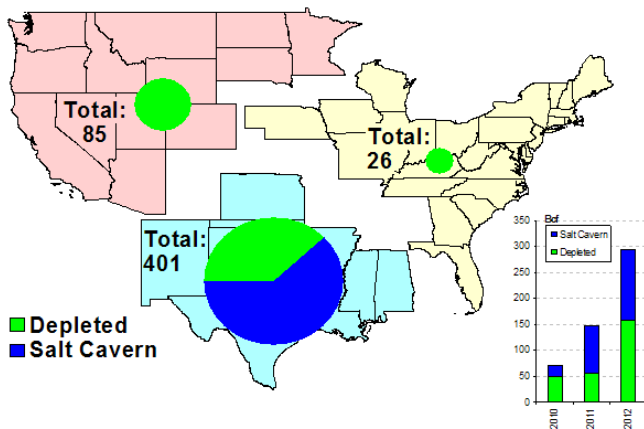
- ▶ ENERGY consists mostly of hydrocarbons (83% of North America's Energy in 2008)
- ▶ In 2008 North America consumed
 - ▶ 27 % of world's oil production
 - ▶ 25 % of world's gas production
 - ▶ 18 % of world's coal production
- ▶ Natural gas is the world's fastest-growing fossil fuel
- ▶ Because USA is the biggest natural gas producer (and consumer) I am going to predict natural gas prices in the USA

Response and endogenous variables

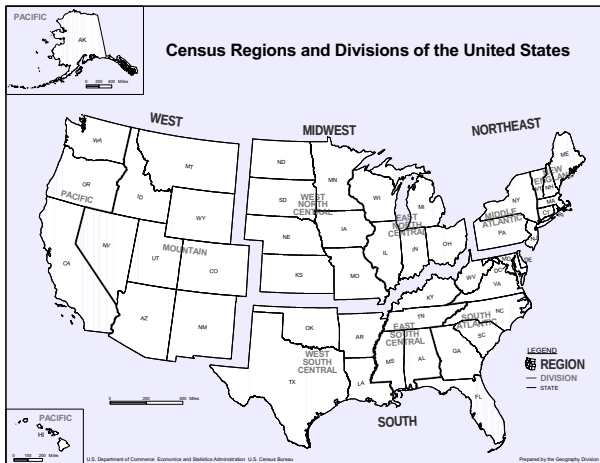
Variable Full name	Unit	Source
Henry Hub Natural Gas Spot Price	Dollars per million British Thermal Unit (\$\textbf{mmBtu})	U.S. Energy Information Administration
West Texas Intermediate Crude Oil Spot Price	Dollars per Barrel (\$\textbf{bbl})	U.S. Energy Information Administration

Exogenous variables

Variable	Unit	Source
Heating degree days	Day	Climate
Cooling degree days	Day	Prediction
Heating degree days deviation	Day	Center
Cooling degree days deviation	Day	(NOAA)
Working gas in underground storage	Billion Cubic Feet (BCF)	U.S. Energy Information
Storage differential gas in underground storage	Billion Cubic Feet (BCF)	Administration
Shut in production	Million Cubic Feet of Gas per Day (MMCF/D)	The Minerals Management Service



Potential natural gas storage capacity additions, 2010-2012
(source:U.S. Energy Information Administration (EIA))



EIA Natural Gas Pipeline regions
(source:U.S. Energy Information Administration (EIA))

Data – Time horizon

- ▶ Number of predicted values $n = 4$.
- ▶ Prediction Set – Observation Numbers: 875 – 878.
- ▶ Prediction Set – Date: 9-May-14 – 30-May-14.

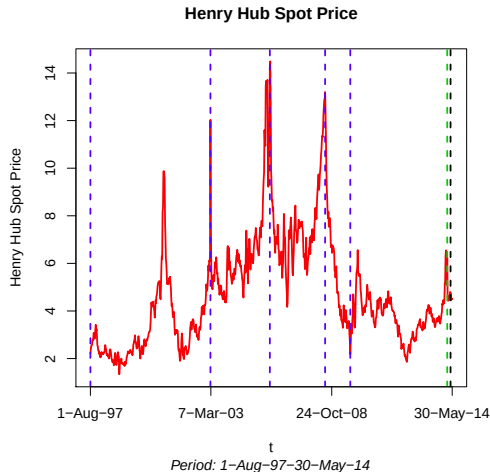
Observation number		Date
Identification set	Ident Set Length	Identification set
1 - 874	874	1-Aug-97 – 2-May-14
292 - 874	583	28-Feb-03 – 2-May-14
436 - 874	439	9-Dec-05 – 2-May-14
570 - 874	305	4-Jul-08 – 2-May-14
631 - 874	244	4-Sep-09 – 2-May-14

Data – Time horizon

- ▶ Number of predicted values $n = 12$.
- ▶ Prediction Set – Observation Number: 867 – 878.
- ▶ Prediction Set – Date: 14-Mar-14 – 30-May-14.

Observation number		Date
Identification set	Ident Set Length	Identification set
1 - 866	866	1-Aug-97 – 7-Mar-14
292 - 866	575	28-Feb-03 – 7-Mar-14
436 - 866	431	9-Dec-05 – 7-Mar-14
570 - 866	297	4-Jul-08 – 7-Mar-14
631 - 866	236	4-Sep-09 – 7-Mar-14

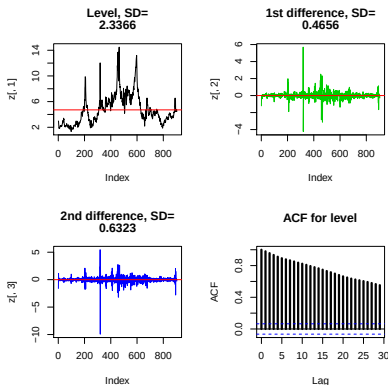
Data – Time horizon



Henry Hub weekly spot prices - time horizon division

Methodology – Stationarity

- ▶ ACF, PACF
- ▶ Unit Root Tests (DF, ADF, PP, KPPS)



Standard deviations for the data, first and second differences, autocorrelation function

- ▶ Autoregressive Moving Average (**ARIMA**)

$$HH_t = \frac{\Theta_q(B)}{\Phi_p(B)\nabla^d} \varepsilon_t \quad (1)$$

- ▶ Autoregressive Moving Average with covariants (**ARIMAX**)

$$HH_t = \frac{\beta}{\Phi_p(B)\nabla^d} X_t + \frac{\Theta_q(B)}{\Phi_p(B)\nabla^d} \varepsilon_t \quad (2)$$

- ▶ Transfer Function Models

$$HH_t = \frac{\beta(B)}{\Phi_p(B)\nabla^d} X_t + \frac{\Theta_q(B)}{\Phi_p(B)\nabla^d} \varepsilon_t \quad (3)$$

- ▶ Error Correction Model (**ECM**)

$$\nabla HH_t = \sum_{i=1}^p \alpha_i \nabla HH_{t-i} + \sum_{i=0}^q \beta_i \nabla WTI_{t-i} + \delta ecm_{t-1} + \varepsilon_t \quad (4)$$

Where:

- ▶ **B** is the backshift operator: $B^d X_t = X_{t-d}$
- ▶ $\nabla X_t = X_t - X_{t-1} = (1 - B)X_t$
- ▶ **HH** – Henry Hub Natural Gas Spot Price and **WTI** – West Texas Intermediate Crude Oil Spot Price.
- ▶ $\Phi_p(B)$ and $\Theta_q(B)$ are autoregressive and moving average polynomials of order p and q respectively and ε_t an error term (white noise process).
 - ▶ $\Phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$
 - ▶ $\Theta_q(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q$
- ▶ X_t is a vector of **endogenous** and **exogenous** variables
 $X_t = [WTI_t, HDD_t, HDDdev_t, CDD_t, CDDdev_t, STOR_PROD_ch_t, HH_shutin_t]$.
- ▶ ecm_t are residuals of the regression $HH_t = \alpha + \beta WTI_t + \varepsilon_t$

Results – ARIMA

ARIMA models for Henry Hub Natural Gas Spot Price,
predicting 4 values

Nr	Model	AIC	BIC	RMSEin	MAPEin	RMSE	MAPE
HH_w; Est. period 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values							
1	ARIMA(5,1,3)	1150.35	1198.08	0.46	5.38	0.26	5.09
2	ARIMA(5,1,2)	1154.89	1197.85	0.46	5.36	0.19	3.82
3	ARIMA(2,1,2)	1156.82	1185.46	0.47	5.33	0.22	4.24
4	ARIMA(4,1,2)	1157.01	1195.19	0.47	5.35	0.18	3.49
HH_w; Est. period 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values							
1	ARIMA(5,1,2)	763.82	803.13	0.46	5.29	0.14	2.89
2	ARIMA(1,1,5)	766.92	801.87	0.46	5.24	0.13	2.67
3	ARIMA(4,1,2)	770.24	805.19	0.46	5.22	0.08	1.70
4	ARIMA(4,1,5)	770.42	818.47	0.46	5.23	0.10	2.17
5	ARIMA(2,1,1)	781.896	803.74	0.47	5.16	0.22	4.34

Results – ARIMA

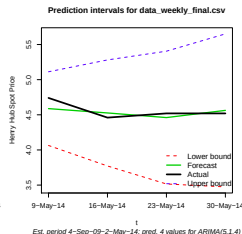
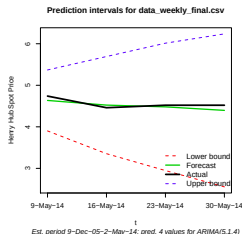
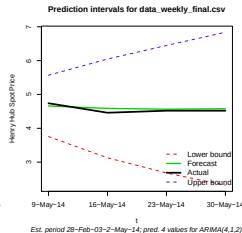
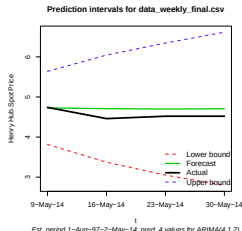
ARIMA models for Henry Hub Natural Gas Spot Price,
predicting 4 values

Nr	Model	AIC	BIC	RMSEin	MAPEin	RMSE	MAPE
HH_w; Est. period 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values							
1	ARIMA(5,1,2)	404.32	441.08	0.37	4.90	0.12	2.46
2	ARIMA(5,1,4)	406.79	451.72	0.37	5.05	0.09	1.81
3	ARIMA(2,1,1)	410.97	431.39	0.38	5.00	0.17	3.27
4	ARIMA(3,1,2)	414.58	443.17	0.38	5.01	0.17	3.20
5	ARIMA(4,1,3)	415.69	452.45	0.38	5.00	0.09	1.57
HH_w; Est. period 4-Jul-08 - 2-May-14 (305 obs); Pred.4 values							
1	ARIMA(3,1,2)	137.35	163.39	0.30	4.83	0.19	3.88
2	ARIMA(2,1,2)	137.63	159.95	0.30	4.86	0.15	2.89
3	ARIMA(4,1,2)	138.90	168.67	0.30	4.84	0.16	3.30
4	ARIMA(2,1,4)	139.51	169.27	0.30	4.83	0.20	3.99
5	ARIMA(0,1,1)	142.59	153.75	0.30	4.79	0.21	4.03

ARIMA models for Henry Hub Natural Gas Spot Price,
predicting 4 values

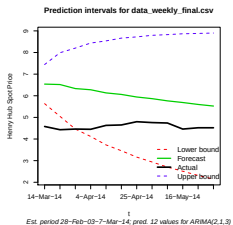
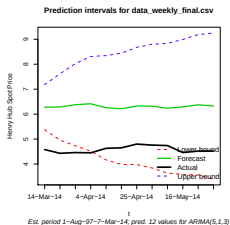
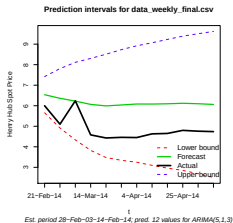
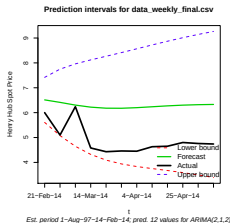
Nr	Model	AIC	BIC	RMSEin	MAPEin	RMSE	MAPE
HH_w; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values							
1	ARIMA(4,1,5)	67.18	105.64	0.26	4.52	0.20	4.16
2	ARIMA(1,1,3)	69.45	90.43	0.27	4.57	0.16	3.02
3	ARIMA(2,1,2)	70.27	91.25	0.27	4.56	0.28	5.39
4	ARIMA(5,1,3)	71.38	106.35	0.27	4.56	0.09	1.73

Results – ARIMA



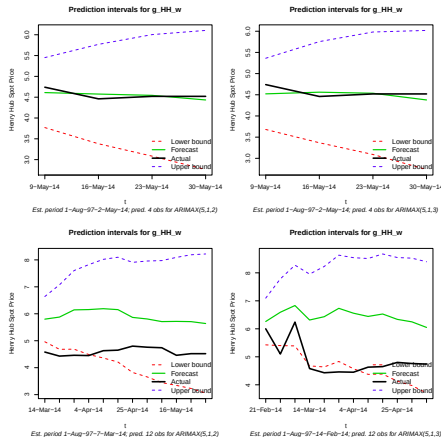
Forecast (n=4) for Henry Hub Natural Gas Spot Price

Results – ARIMA



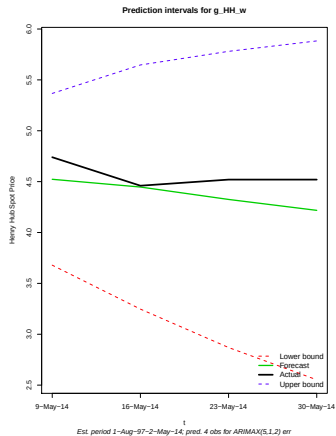
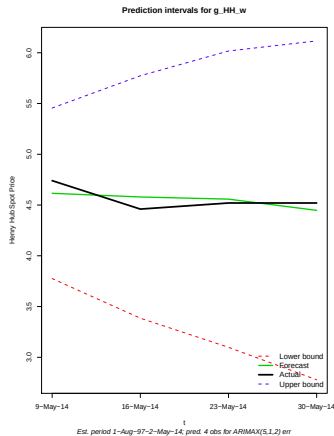
Forecast (n=12) for Henry Hub Natural Gas Spot Price

Results – ARIMAX



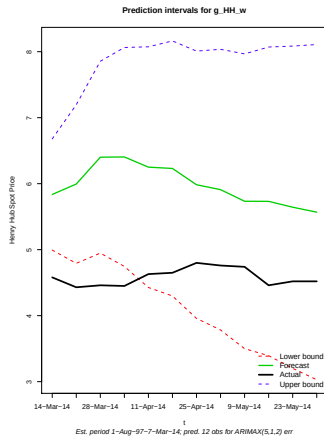
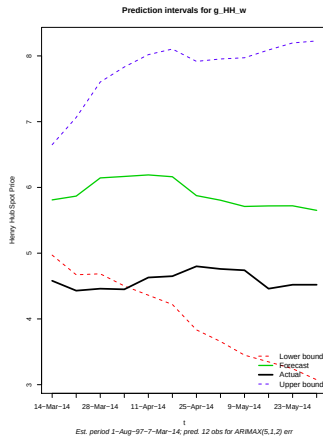
Forecast (n=4 and n=12) for Henry Hub Natural Gas Spot Price
ARIMAX models for local and global data

Results – Transfer Function Models



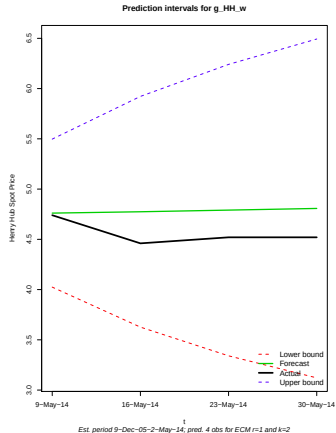
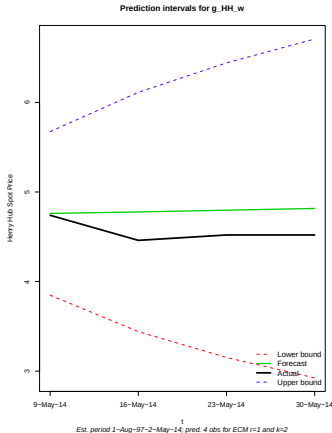
Forecast (n=4) for Henry Hub Natural Gas Spot Price –
Transfer Function models (with 2 lags) for local and global data

Results – Transfer Function Models



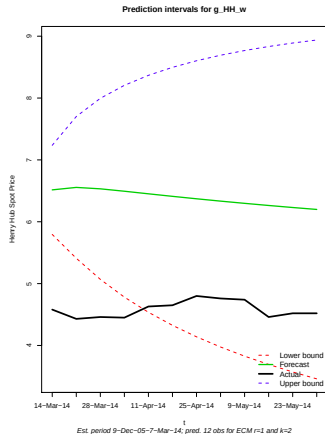
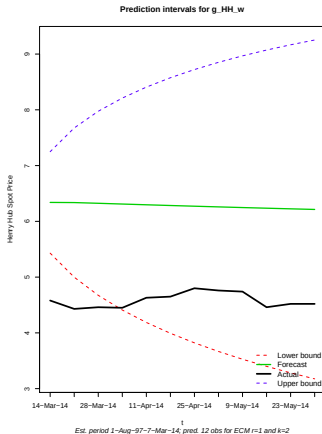
Forecast (n=12) for Henry Hub Natural Gas Spot Price –
Transfer Function models (with 2 lags) for local and global data

Results – ECM



Forecast (n=4) for Henry Hub Natural Gas Spot Price – ECM

Results – ECM



Forecast (n=12) for Henry Hub Natural Gas Spot Price – ECM

Results – Model comparison

Model comparison for Henry Hub Natural Gas Spot Price – 1-874, predicting 4 values

Nr	Model	Param	RMSEin	MAPEin	RMSE	MAPE
Estimation comp; 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values						
1	Trans Fun local	ARIMAX(5,1,2)	0.43	5.52	0.10	1.89
2	ARIMAX local	ARIMAX(5,1,2)	0.43	5.52	0.10	1.94
3	Trans fun global 2	ARIMAX(1,1,1)	0.44	5.39	0.12	2.39
4	ARIMAX global	ARIMAX(5,1,3)	0.43	5.45	0.14	2.60
5	Trans Fun local	ARIMAX(5,1,3)	0.43	5.51	0.16	3.21
6	ARIMAX local	ARIMAX(5,1,3)	0.43	5.52	0.16	3.29
7	ARIMA	ARIMA(4,1,2)	0.47	5.35	0.18	3.49
8	Trans fun global 2	ARIMAX(5,1,2)	0.43	5.45	0.21	3.97
9	ARIMA	ARIMA(2,1,2)	0.47	5.33	0.22	4.24
10	ECM	ECM r=1 k=2	0.47	5.37	0.26	5.06

Results – Model comparison

Model comparison for Henry Hub Natural Gas Spot Price – 1-866, predicting 12 values

Nr	Model	Param	RMSEin	MAPEin	RMSE	MAPE
Estimation comp; Est. per. 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values						
1	Trans Fun local 1	ARIMAX(5,1,3)	0.43	5.49	1.22	26.26
2	ARIMAX local	ARIMAX(5,1,3)	0.43	5.49	1.24	26.77
3	Trans Fun local 1	ARIMAX(5,1,2)	0.43	5.49	1.33	28.63
4	ARIMAX local	ARIMAX(5,1,2)	0.43	5.49	1.34	28.79
5	ARIMAX global	ARIMAX(5,1,3)	0.43	5.41	1.35	28.89
6	Trans fun global 1	ARIMAX(1,1,1)	0.44	5.36	1.38	29.48
7	ARIMAX global	ARIMAX(5,1,2)	0.43	5.43	1.43	30.46
8	ARIMA	ARIMA(2,1,2)	0.47	5.33	1.52	32.88
9	ARIMA	ARIMA(4,1,3)	0.46	5.35	1.68	36.67
10	ECM	ECM r=1 k=2	0.46	5.36	1.70	37.10

Results – Model comparison

Model comparison for Henry Hub Natural Gas Spot Price – 570-874, predicting 4 values

Nr	Model	Param	RMSEin	MAPEin	RMSE	MAPE
Est. comp; Est. per. 4-Sep-09 - 2-May-14 (305 obs); Pred.4 values						
1	ECM	ECM $r=1$ $k=2$	0.28	4.80	0.08	1.67
2	Trans fun global 1	ARIMAX(3,1,0)	0.28	4.73	0.10	1.74
3	Trans fun global 1	ARIMAX(1,1,1)	0.28	4.65	0.10	1.78
4	ARIMAX global	ARIMAX(3,1,0)	0.28	4.71	0.10	1.83
5	ARIMAX global	ARIMAX(2,1,0)	0.28	4.71	0.11	1.84
6	Trans fun global 2	ARIMAX(5,1,0)	0.28	4.69	0.11	2.15
7	Trans fun global 2	ARIMAX(3,1,2)	0.28	4.68	0.12	1.95
8	Trans fun global 2	ARIMAX(2,1,3)	0.28	4.66	0.12	2.22
9	ARIMAX global	ARIMAX(2,1,3)	0.28	4.68	0.12	2.47
10	ARIMAX global	ARIMAX(5,1,3)	0.28	4.67	0.12	1.98

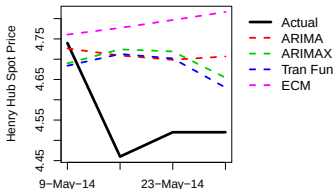
Results – Model comparison

Model comparison for Henry Hub Natural Gas Spot Price – 570-866, predicting 12 values

Nr	Model	Param	RMSEin	MAPEin	RMSE	MAPE
Est. comp; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred. 12 values						
1	ECM	ECM $r=1$ $k=2$	0.27	4.71	1.02	19.43
2	ARIMAX local	ARIMAX(1,1,1)	0.26	4.55	1.43	30.82
3	Trans Fun local 1	ARIMAX(3,1,2)	0.26	4.52	1.45	31.22
4	ARIMAX local	ARIMAX(3,1,0)	0.26	4.55	1.46	31.44
5	Trans Fun local 1	ARIMAX(1,1,1)	0.26	4.54	1.46	31.66
6	ARIMAX local	ARIMAX(2,1,2)	0.26	4.52	1.50	32.51
7	Trans Fun local 2	ARIMAX(3,1,0)	0.26	4.52	1.52	32.86
8	Trans fun global 1	ARIMAX(3,1,2)	0.26	4.53	1.54	32.87
9	Trans fun global 1	ARIMAX(2,1,2)	0.26	4.53	1.57	33.64
10	ARIMA	ARIMA(3,1,5)	0.27	4.65	1.86	40.33

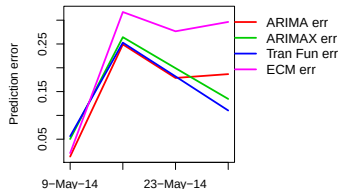
Results – Model comparison

Predictions for g_HH_w



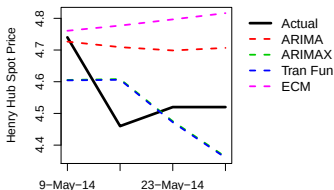
Est. per 1-Aug-97-2-May-14 (nr 1-874) ; pred. 4 val

Prediction errors for g_HH_w



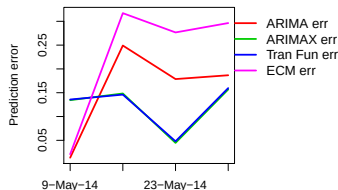
Est. per 1-Aug-97-2-May-14 (nr 1-874) ; pred. 4 val

Predictions for g_HH_w



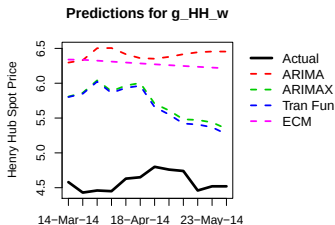
Est. per 1-Aug-97-2-May-14 (nr 1-874) ; pred. 4 val

Prediction errors for g_HH_w

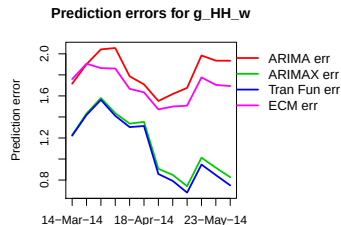


Est. per 1-Aug-97-2-May-14 (nr 1-874) ; pred. 4 val

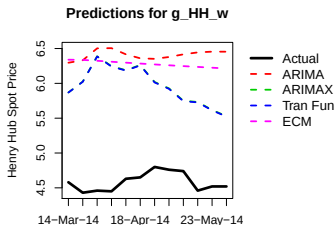
Results – Model comparison



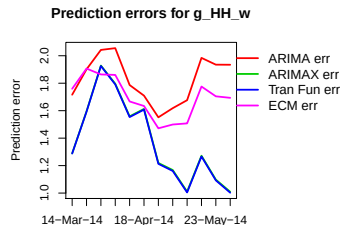
t
Est. per 1-Aug-97-7-Mar-14 (nr 1-866) ; pred. 12 val



t
Est. per 1-Aug-97-7-Mar-14 (nr 1-866) ; pred. 12 val



t
Est. per 1-Aug-97-7-Mar-14 (nr 1-866) ; pred. 12 val



t
Est. per 1-Aug-97-7-Mar-14 (nr 1-866) ; pred. 12 val

Results – Model comparison

Prediction comparison for Henry Hub Natural Gas Spot Price – 1-874, predicting 4 values

Estimation period 1-Aug-97-2-May-14 (obs nr 1-874) ; pred.4 val					
Model Pred	ACTUAL	ARIMA (4,1,2)	ARIMAX local	Trans Fun 2 lags	ECM r=1 k=2
1	4.74	4.73	4.69	4.68	4.76
2	4.46	4.71	4.72	4.71	4.78
3	4.52	4.70	4.72	4.70	4.80
4	4.52	4.71	4.65	4.63	4.82
Err nr	ACTUAL	ARIMA err	ARIMAX err	Trans Fun err	ECM err
1		0.01	0.05	0.06	0.02
2		0.25	0.26	0.25	0.32
3		0.18	0.20	0.18	0.28
4		0.19	0.13	0.11	0.30

Results – Model comparison

Prediction comparison for Henry Hub Natural Gas Spot Price –
local data, 1-866, predicting 12 values

Estimation period 1-Aug-97-7-Mar-14 (obs nr 1-866) ; pred.12 val					
Model Pred	ACTUAL	ARIMA (4,1,2)	ARIMAX local	Trans Fun 2 lags	ECM r=1 k=2
1	4.58	6.30	5.81	5.80	6.34
2	4.43	6.33	5.86	5.85	6.34
3	4.46	6.50	6.04	6.02	6.32
4	4.45	6.51	5.89	5.86	6.31
5	4.63	6.42	5.97	5.93	6.30
6	4.65	6.36	6.00	5.96	6.28
7	4.8	6.35	5.71	5.66	6.27
8	4.76	6.38	5.61	5.55	6.26
9	4.74	6.42	5.48	5.42	6.25
10	4.46	6.44	5.47	5.41	6.24
11	4.52	6.46	5.44	5.37	6.22
12	4.52	6.45	5.35	5.27	6.21

Results – Model comparison

Prediction comparison for Henry Hub Natural Gas Spot Price –
local data, 1-866, predicting 12 values

Estimation period 1-Aug-97-7-Mar-14 (obs nr 1-866) ; pred.12 val				
Err nr	ARIMA err	ARIMAX err	Trans fun err	ECM err
1	1.72	1.23	1.22	1.76
2	1.90	1.43	1.42	1.91
3	2.04	1.58	1.56	1.86
4	2.06	1.44	1.41	1.86
5	1.79	1.34	1.30	1.67
6	1.71	1.35	1.31	1.63
7	1.55	0.91	0.86	1.47
8	1.62	0.85	0.79	1.50
9	1.68	0.74	0.68	1.51
10	1.98	1.01	0.95	1.78
11	1.94	0.92	0.85	1.70
12	1.93	0.83	0.75	1.69

Conclusions

- ▶ The prediction results do not depend solely on the applied method but also on a multitude of other factors (time horizon, variables, data etc.).
- ▶ ARIMA and ECM models' forecasts are similar.
- ▶ ARIMAX and Transfer Function models' forecasts are following also a common pattern.
- ▶ Transfer Function, ARIMAX and ARIMA are better for longer estimation time horizon.
- ▶ ECM are better for shorter estimation time horizon.
- ▶ Local data are slightly better in predicting long-time horizons than global data.

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Thank you for your attention!